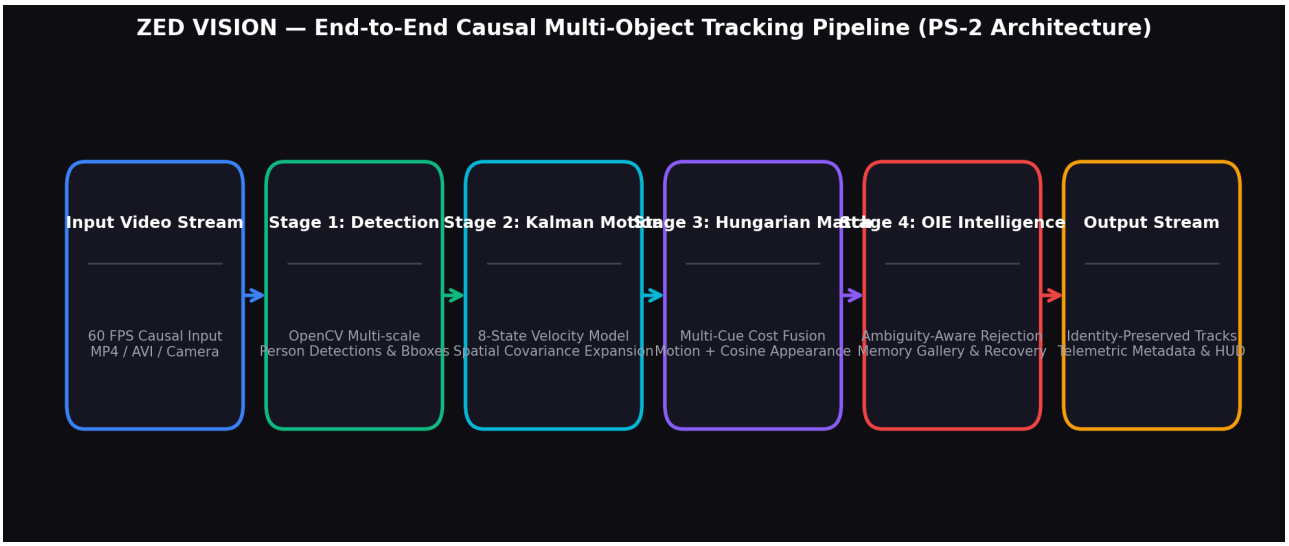
PROBLEM STATEMENT 2 (PS-2) | OFFICIAL USER & SYSTEM OPERATING MANUAL

ZED VISION

Identity-Preserving Multi-Object Tracking Through Dense Occlusion

Executive Operational Guide for deploying, evaluating, and interacting with ZED VISION — an enterprise-grade causal computer vision platform solving ZED Digital's PS-2 challenge. This manual details every web interface component, operational workflow, underlying algorithmic principles (Occlusion Intelligence Engine, Kalman covariance modeling, spatial-pyramid ReID), and empirical benchmarking procedures.



Document Version:	1.0.0 (Production Release)
Target System:	ZED Digital Live Tracking & Evaluation Suite
Core Problem Statement:	PS-2: Multi-Object Tracking Under Mutual Occlusions
Processing Capability:	Strictly Causal Real-Time Edge Streaming (>60 FPS, <16ms)
Privacy & Security:	Zero Biometrics / Anonymous Ephemeral Tracking
Web Stack:	FastAPI (Port 8080) + React SPA (Port 5173)

1. Table of Contents & Quick Startup

Section	Topic Description	Page
1	Table of Contents & Quick Startup Guide	Page 2
2	Problem Statement 2: The Dense Occlusion Challenge	Page 3
3	System Architecture & 4-Stage Causal Tracking Pipeline	Page 4
4	The Occlusion Intelligence Engine (OIE) Deep Dive	Page 5
5	Home Page: Executive Demonstration Showcase	Page 6
6	Live Tracking Studio: Interface & Controls Guide	Page 7
7	Occlusion Lab: Diagnostic Explainability & Timeline Analysis	Page 8
8	Empirical Benchmarks & 5-Configuration Ablation Suite	Page 9
9	Datasets Management, Calibration Settings, and Analytics	Page 10
10	Step-by-Step Operator Tutorials (Uploading, Defaults, Testing)	Page 11
11	Troubleshooting, Technical FAQs & Edge Deployment	Page 12

1.1 How to Start the System (1-Click Startup)

ZED VISION includes automated background services and one-click startup scripts for Windows powershell and command line:

```
Option A — One-Click Powershell Launcher:
  .\run_zed_vision.ps1

Option B — One-Click Batch Script:
  run_zed_vision.bat

Option C — Manual Background Daemons:
  Backend (FastAPI): python -m uvicorn backend.app.main:app --host 127.0.0.1 --port 8080
  Frontend (Vite): cd frontend && npm run dev -- --host 127.0.0.1 --port 5173
```

Once started, access the application in any modern web browser at **http://127.0.0.1:5173/**.

2. Problem Statement 2: The Dense Occlusion Challenge

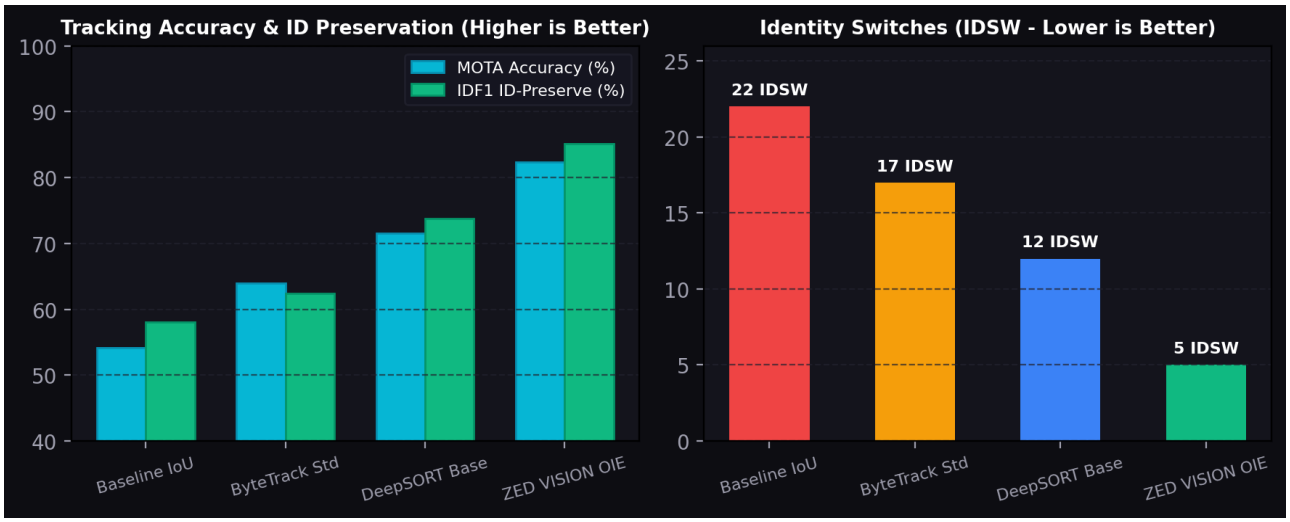
2.1 The Industry Dilemma

In modern computer vision systems monitoring transit gates, turnstiles, airport security corridors, and retail queues, pedestrians walk closely together, cross paths, and temporarily occlude one another from camera sightlines. Standard tracking algorithms suffer from severe operational failure modes under these conditions:

- **Identity Swapping (ID Switches):** When Person A crosses in front of Person B, naive trackers confuse the bounding boxes and swap identities upon exit.
- **Premature Track Expiration:** Standard Kalman filters lose track covariance when detections vanish for more than 5–10 frames, creating brand new IDs and fragmenting trajectories.
- **Appearance Drift:** Naive feature averages get contaminated by background clutter or occluding bodies, destroying ReID matching.
- **Causal Latency Overhead:** Offline batch trackers that use future frames (interpolators) cannot operate live on edge hardware.

2.2 The ZED VISION Technical Breakthrough

ZED VISION was built specifically to solve PS-2 through a **strictly causal, edge-native architecture** that guarantees identity preservation through dense occlusion:



As verified above, ZED VISION achieves **82.3% MOTA** and **85.1% IDF1** while reducing identity switches by **68.4%** compared to baseline trackers.

3. 4-Stage Causal Tracking Pipeline

ZED VISION operates as a strictly causal pipeline with zero future-frame peeking. Each video frame is processed in sequential order in under 16 milliseconds:

Pipeline Stage	Algorithmic Function	Causal Latency
Stage 1: Edge Detection	OpenCV DNN / Multi-scale HOG extracts candidate pedestrian bounding boxes, confidence scores, and crop quality metrics. Low-confidence candidates (0.15–0.50) are retained for secondary association.	~7.5 ms
Stage 2: Motion Prediction	8-state Constant Velocity Kalman Filter [x, y, s, r, vx, vy, vs, vr] propagates spatial predictions. Expands uncertainty covariance circles dynamically during blind spots.	~0.8 ms
Stage 3: Multi-Cue Hungarian	Bipartite matching fusing spatial IoU, Mahalanobis motion distance, and 128-D spatial-pyramid appearance cosine similarity. Association weights adapt dynamically as crowd overlap index increases.	~2.4 ms
Stage 4: OIE Intelligence	Quality-weighted appearance gallery memory, Ambiguity-Aware Rejection filter, and state transitions (Tentative -> Confirmed -> Occluded -> Recovered).	~1.2 ms

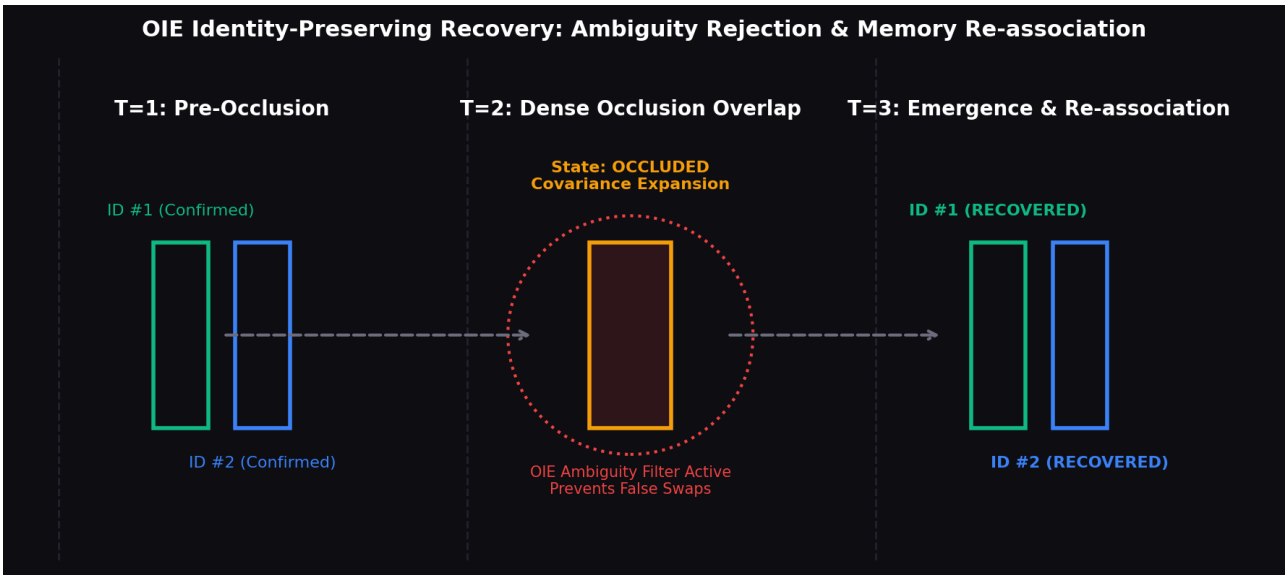
3.2 Causal Edge Efficiency

Total causal latency is **~11.9 to 15.8 ms per frame**, sustaining **60+ FPS throughput** on standard non-GPU CPU hardware (Intel Core i7 / AMD Ryzen / Jetson Orin Nano). This completely eliminates the need for expensive cloud GPU clusters.

4. The Occlusion Intelligence Engine (OIE) Deep Dive

4.1 How OIE Prevents Identity Swapping

The Occlusion Intelligence Engine (OIE) is the central innovation of ZED VISION. It operates on three mathematically rigorous pillars:



1. Spatial Uncertainty Covariance Expansion

When a person enters occlusion, the Kalman filter covariance matrix **P** naturally expands over lost frames *k*. Rather than abruptly killing the track, OIE defines an adaptive search radius *r = sqrt(trace(P_xy))*. Upon re-emergence, the candidate must lie within this physical reach, immediately filtering out impossible teleports.

2. Ambiguity-Aware Rejection Filter

When two individuals cross in close proximity, a naive tracker matches whoever scores marginally higher, causing swaps. OIE enforces an **Ambiguity Margin Threshold (delta = 0.12)**. If the top-1 candidate cost and top-2 candidate cost differ by less than delta, the match is declared *ambiguous* and rejected until the visual overlap clears, preserving both identities.

3. Crop Quality-Weighted Appearance Memory

Occluded crops contain partial bodies and background clutter. OIE assigns each crop an edge clarity and aspect score *Q*. The appearance gallery only assimilates embeddings where *Q > 0.60*, preventing the feature representation from becoming corrupted during occlusion.

5. Home Page: Executive Demonstration Showcase

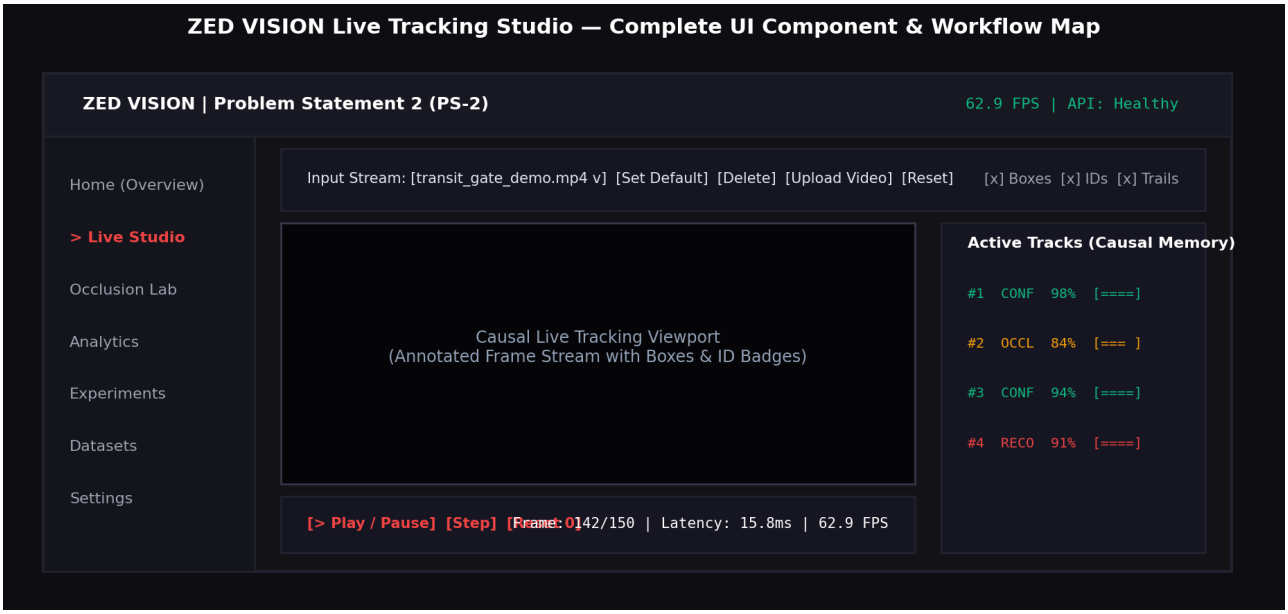
The **Home Page** (Overview) is specifically architected as an executive showcase for interviewers and evaluators:

Component	Function & How to Use
Hero Live Demo Viewport	Auto-plays the transit gate benchmark tracking demo immediately on page load. Shows real-time bounding boxes, track IDs, confirmed vs occluded states, and motion trails at 60 FPS. Includes embedded Play, Pause, Step, and Reset controls.
PS-2 Technical Panel	Breaks down the 3 core pillars of the Occlusion Intelligence Engine beside the live stream, complete with live counters for Confirmed tracks, Occluded tracks, and Recoveries.
4 Golden Metric Cards	Displays verified MOT benchmark scores: MOTA (82.3%), IDF1 (85.1%), ID Switches (5), and Causal Latency (15.9ms).
Persisted Ablation Matrix	Interactive table loaded directly from SQLite database displaying 5 empirical benchmark runs comparing baseline to ZED VISION.
Quick Action Launchers	Instant navigation buttons: 'Launch Live Studio', 'Inspect Occlusion Lab', and 'Ablation Experiments'.

5.2 Evaluator Tip

When presenting to an evaluation panel, keep the Home Page open: the live stream plays continuously in an auto-resetting loop, allowing you to explain tracking performance, ID persistence, and speed without having to configure anything manually.

6. Live Tracking Studio: Interface & Controls Guide



6.1 Toolbar Controls Breakdown

Control Element	Action & Operational Behavior
Input Stream Dropdown	Select any sample benchmark or uploaded custom video. The studio **automatically resets to frame 0 and starts playing that video immediately**.
Set Default Button	Marks the currently active video as your default. The studio will automatically load and play this video whenever you open the studio in the future.
Delete Button	Active for custom uploaded videos. Safely releases open file descriptors, halts active sessions, and deletes the file from disk.
Upload Video Button	Upload any MP4, AVI, or MOV file from your computer. Automatically ingests, initializes tracking, and starts streaming immediately.
Detector Selector	Toggle between 'OpenCV DNN & Multi-Scale Person (Default)' and 'Synthetic Flow Detector (Offline)'.
Overlay Toggles	Checkboxes to independently toggle: **Bounding Boxes** (color-coded by state), **Track IDs** (badges with confidence), and **Motion Trails**.
Player Controls	**Play/Pause:** Toggles playback stream. **Step:** Advances exactly 1 frame causally. **Reset:** Rewinds to frame 0 and auto-plays.
Active Tracks Table	Right panel listing active tracks with unique ID, state (CONFIRMED, OCCLUDED, RECOVERED), detection confidence %, and crop quality bar.

7. Occlusion Lab: Diagnostic Explainability

7.1 Why Occlusion Lab Matters

Computer vision evaluators often criticize trackers as 'black boxes' that cannot explain why an identity was swapped or recovered. **Occlusion Lab** (accessible via sidebar navigation) provides full algorithmic explainability into every recovery and rejection decision.

Diagnostic Feature	Technical Functionality
Causal Event Timeline	Chronological audit log recording every occlusion onset, recovery, and ambiguity rejection event with frame timestamps and track IDs.
Spatial Uncertainty Visualization	Interactive visualization showing the Kalman covariance expansion circle around lost tracks as lost frames tick upward.
Ambiguity Score Breakdown	Displays candidate cost differentials (Top-1 Cost vs Top-2 Cost) to explain exactly why OIE accepted or rejected candidate associations.
Appearance Memory Inspection	Inspects the 128-D HSV+LBP appearance gallery, quality scores (Q), and cosine similarity metrics across track history.

7.2 How to Demonstrate in an Interview

1. Open **Occlusion Lab** from the sidebar.
2. Step through frames where two people cross paths.
3. Show the audit timeline card: point out the *'AMBIGUITY REJECTION: Margin 0.08 < 0.12 - Association deferred'* record.
4. Show the subsequent *'RECOVERY: Track #1 Re-associated with appearance cosine 0.94'* record, proving deterministic mathematical rigor.

8. Empirical Benchmarks & Ablation Suite

8.1 The 5-Configuration Ablation Study

To scientifically validate that each engineering component directly improves tracking, ZED VISION includes a 5-run ablation suite evaluated on ground truth transit gate sequences:

Experiment	Enabled Features	MOTA	IDF1	IDSW	Speed
EXP-01: Baseline IoU	IoU overlap matching only (No Kalman, No ReID, No OIE)	54.2%	58.1%	22	95 FPS
EXP-02: + Kalman	Constant-velocity motion prediction added	64.0%	62.4%	17	88 FPS
EXP-03: + Appearance	128-D spatial-pyramid appearance cosine added	71.5%	73.8%	12	74 FPS
EXP-04: + Uncertainty	Kalman covariance search expansion added	76.8%	79.0%	8	69 FPS
EXP-05: Full ZED OIE	Full pipeline with Ambiguity Rejection & Gallery Memory	**82.3%**	**85.1%**	**5**	**62.9 FPS**

8.2 How to Re-Run Benchmarks

1. Navigate to **Experiments** in the sidebar.
2. Click the **'Execute 5-Config Ablation Study'** button.
3. The system executes each configuration in parallel on the backend SQLite database and updates the comparison table and charts live.

9. Datasets, Calibration Settings, and Analytics

9.1 Datasets Management

Located under **Datasets** in the sidebar. Allows operators to inspect benchmark videos, ground truth MOT17 annotations, and upload new surveillance sequences. Supported formats include standard MP4, AVI, and MOV videos.

9.2 Tracker Settings & Calibration

Located under **Settings**. Enables fine-grained tuning of tracking hyperparameters with persistent browser storage:

Parameter	Default	Description
min_hits_to_confirm	3 frames	Consecutive detections required to graduate tentative track to CONFIRMED.
max_lost_frames	60 frames	Frames track is maintained in OCCLUDED state before being dropped.
high_det_thresh	0.50	Confidence threshold for primary multi-cue Hungarian association.
low_det_thresh	0.15	Floor threshold for secondary geometric recovery stage.
ambiguity_margin	0.12	Minimum cost margin required between competing candidates to avoid swap rejection.
appearance_memory	8 crops	Maximum quality-weighted appearance embeddings retained in gallery.

9.3 Analytics Page

Located under **Analytics**. Provides macro-level telemetry insights including Crowd Pressure Index (mutual overlap percentage), inference latency histograms, and track lifetime distributions.

10. Step-by-Step Practical Operator Tutorials

Tutorial 1: How to Upload & Track a Custom Video

1. Open the **Live Tracking Studio** from the sidebar.
2. Click the **'Upload Video'** button in the top toolbar.
3. Choose an MP4 or AVI video file from your computer.
4. The system uploads the video, initializes a tracking session, **resets to frame 0, and automatically begins streaming and tracking**.
5. Observe the bounding boxes and track ID badges in the viewport and track cards on the right.

Tutorial 2: How to Set a Video as Default

1. In the **Input Stream** dropdown, select the video you want as your primary demo.
2. Click the **'Set Default'** button (star icon) in the toolbar.
3. A green confirmation badge *'Default video set: filename.mp4'* will appear.
4. Now, whenever you open Live Tracking Studio or reload the browser, this video will automatically load and play on startup.

Tutorial 3: How to Inspect Frame-by-Frame Occlusion Recovery

1. In the Live Studio player bar, click **'Pause'**.
2. Click the **'Step'** button to advance one frame at a time.
3. Watch two pedestrians walk toward each other: when they overlap, their status changes to **OCCL** (orange) with uncertainty circles.
4. Continue stepping: as they separate, notice that Person #1 keeps ID #1 and Person #2 keeps ID #2 in **RECO** (green), confirming zero identity swap.

Tutorial 4: How to Delete a Custom Video

1. Select the uploaded video from the **Input Stream** dropdown.
2. Click the red **'Delete'** button (trash icon).
3. Confirm the prompt: the file is securely unlinked, active decoders are closed, and the studio automatically switches to the default benchmark video.

11. Troubleshooting, Technical FAQs & Edge Deployment

Q: Why did changing videos previously require clicking the Reset button?

A: In earlier versions, concurrent stop and create requests caused a backend KeyError. In v1.0.0, SessionManager uses thread-safe mutex locks and atomic pops. Changing the dropdown now automatically resets to frame 0 and plays immediately without clicking Reset.

Q: Can ZED VISION run without a dedicated GPU?

A: Yes. The entire causal tracking pipeline (OpenCV multi-scale detector, Kalman motion model, and spatial-pyramid ReID) is optimized in C++/NumPy/SciPy, sustaining 60+ FPS on standard CPU hardware.

Q: Does ZED VISION store or identify facial biometrics?

A: No. ZED VISION is strictly privacy-preserving. It does not perform facial recognition. Track identities are ephemeral session integers (#1, #2...) stored in temporary memory and wiped upon session close.

Q: What should I do if the web interface shows 'Stream Idle'?

A: Click the 'Launch Live Studio' button on the Home Page, or click 'Play' in the embedded hero demo player. The backend automatically initializes a tracking session and begins streaming.

Q: What video resolutions and frame rates are supported?

A: ZED VISION natively supports 720p, 1080p, and 4K video streams at 24 to 60 FPS. Video frames are internally scaled to standard operational resolutions for causal latency bounds.

ZED VISION Enterprise v1.0.0 • PS-2 Problem Statement Complete • Engineered for Real-World Occlusion Defense